

Smart Farming Agent – Multi-Agent AI System Blueprint

Problem Statement

Build an AI-powered Smart Farming Agent using LangFlow, IBM watsonx.ai, and IBM Granite Models. The system provides personalized agricultural advice, weather intelligence, pest and disease detection, market insights, and farm cost optimization.

Core Objectives

1. Agricultural Knowledge Retrieval (RAG) for crop best practices, government advisories, and farming guidelines.
2. Weather & Soil Intelligence for irrigation planning and climate-aware farming decisions.
3. Pest & Disease Detection using image uploads and multimodal AI analysis.
4. Personalized Farming Recommendations based on crop, soil, location, and farmer profile.
5. Market & Cost Optimization for maximizing profitability and reducing farming expenses.
6. Multi-Agent Architecture for specialized decision-making and collaboration.

System Architecture

Farmer Input Layer → Agent Orchestrator → Multi-Agent Layer → RAG Layer → IBM watsonx.ai Granite Models → Recommendation & Insights Layer.

Multi-Agent Layer

- Crop Advisory Agent: Provides crop-specific recommendations, fertilizers, seeds, and harvesting guidance.
- Weather & Irrigation Agent: Analyzes weather forecasts and irrigation requirements.
- Pest & Disease Detection Agent: Identifies diseases and pests from crop images and suggests treatments.
- Market Insights Agent: Provides mandi prices, demand forecasting, and selling recommendations.
- Cost Optimization Agent: Calculates ROI, optimizes inputs, and improves profitability.
- Government Scheme Agent: Recommends subsidies, insurance, and farmer welfare schemes.

RAG Pipeline

Document Loader → Text Splitter → Granite Embeddings → Vector Store (Milvus/Chroma/Pinecone) → Retriever → Context Injection. Sources include ICAR Guidelines, Government Advisories, Agricultural Research Papers, Soil Health Reports, Weather Bulletins, and Crop Management Documents.

IBM Technology Stack

- LangFlow for orchestration and agent workflows.
- IBM watsonx.ai for model deployment and inference.
- IBM Granite Models for reasoning and advisory generation.
- IBM Granite Vision for pest and disease image analysis.
- IBM Granite Embeddings for semantic retrieval.
- Vector Database: Milvus, Chroma, or Pinecone.

Input Channels

- Text-based farmer queries.
- Crop image uploads for disease detection.
- Voice-based farmer interaction in local languages.
- Weather, soil, and farm profile data.
- Market and commodity price feeds.

Output Features

- Personalized crop advisory.
- Fertilizer and seed recommendations.
- Irrigation scheduling and weather alerts.
- Pest and disease diagnosis reports.
- Market price forecasts and selling strategies.
- Cost optimization and profitability analysis.

Dashboard Metrics

Crop Health Score, Soil Moisture Level, Rainfall Forecast, Irrigation Requirement, Pest Risk Index, Market Price Trends, Yield Prediction, Cost Analysis, and Profit Forecast.

Expected Benefits

- Improved crop yield and productivity.
- Early detection of pests and diseases.
- Reduced water and fertilizer wastage.
- Better market timing and income generation.
- Climate-resilient and sustainable farming practices.
- Data-driven agricultural decision-making.